# Md Ashraful Hossen Akash

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## RESEARCH INTERESTS

Computer Vision, Medical Imaging, Healthcare AI, Federated Learning.

## EDUCATION

**B.Sc. in Computer Science & Engineering** Mar 2022 - July 2026
Rajshahi University of Engineering & Technology
**CGPA**: 3.25/4.00 **(3.68 in last 61.75 Credits)**

## PUBLICATIONS & PREPRINTS

### Preprints & Manuscripts Under Review

**[P1]** Abdullah Al Mamun, Md. Nasif Osman Khansur, **Md Ashraful Hossen Akash**, Md. Kishor Morol, Tze Hui Liew. "Beyond Accuracy: Quantifying Pulmonary Attribution in Anatomy-Guided Chest X-Ray Classification Under Domain Shift." *Under review at Scientific Reports.*
[Paper] [arXiv] [Code]

**[P2]** **Md Ashraful Hossen Akash**, Shyla Afroge, Abdullah Al Mamun, Md. Kishor Morol, Tze Hui Liew. "TRIAGE: Risk-Controlled Pseudo-Label Admission for Annotation-Efficient Semi-Supervised Retinal OCT Classification." *Under review at Scientific Reports.*
[Paper] [arXiv] [Code]

**[P3]** Md Abdur Rahaman, **Md Ashraful Hossen Akash**, Ayman Tahmid, Farjana Parvin. "MeLA-Light: Melanoma Classification Using Multi-Stage DCT-Based Frequency-Channel Attention Fusion on RepViT." *Under review at 7th ICRPSET 2026.*
[Paper]

**[P4]** Md. Tanvir Kabir Akon, A. F. M. Minhazur Rahman, **Md Ashraful Hossen Akash**. "Compact Calibrated Prototypical Networks for Cross-Domain Few-Shot Plant Disease Classification." *Under review at 29th ICCIT 2026.*
[Paper] [Code]

### Peer-Reviewed Conference Proceedings

**[C1]** **Md Ashraful Hossen Akash**, Shyla Afroge, Md. Asif Sajeed, Md Barkullah, Abul Khayer. "An Explainable Deep Learning Framework for Multi-Class Alzheimer's Diagnosis with Multi-Scale CNN." *in 2nd QPAIN 2026.*
[Paper] [DOI] [Code]

**[C2]** Md. Asif Sajeed, Shyla Afroge, **Md Ashraful Hossen Akash**, Abul Khayer, Abdullah Al Mamun. "Memory-Efficient Deep Learning Architectures for Water Body Segmentation in Satellite Imagery: A Comparative Analysis and Ensemble." *in 28th IEEE ICCIT 2025.*
[Paper] [DOI] [Code]

**[C3]** Abul Khayer, Shyla Afroge, Abdullah Al Mamun, Md. Asif Sajeed, **Md Ashraful Hossen Akash**. "Transformer-Based Multi-Label Disease Prediction from Radiology Reports." *in 28th IEEE ICCIT 2025.*
[Paper] [DOI] [Code]

## EXPERIENCE

**Research Intern** Aug 2026 – Present
Elite Research Lab LLC *New York, USA*
Supervised by Kishor Morol

- Conducting research on multimodal architectures and federated learning for healthcare AI applications.

**Research Assistant** June 2026 – Present
Arché Intelligence Lab *CSE, AIUB*
Supervised by Mahamodul Hasan Mahadi

- Spearheaded deep learning and LLM research for neuro-oncological imaging, focusing on MRI-based brain tumor classification, semantic segmentation (tumor boundary delineation), and automated grading.

**Student Researcher** May 2024 – Aug 2026
Supervised by Shyla Afroge *CSE, RUET*

- Conducted research in deep learning for medical imaging, satellite imagery, and clinical text analysis, including data preprocessing, model development, and experimental evaluation.
- Contributed to three published international conference papers (ICCIT 2025, QPAIN 2026) involving multi-scale CNN architectures, transformer-based NLP models, and ensemble deep learning methods.

**Software Engineer Intern** Feb 2025 – May 2025
Code Studio Software Company *Rajshahi, Bangladesh*

- Applied SDLC workflows, relational DB schema design, and SQL (CRUD, JOINs, aggregates).
- Built full-stack CRUD/blog apps (Node.js, Django, React, Next.js) with SQL/NoSQL DBs (MySQL, MongoDB, PostgreSQL), REST APIs, auth, middleware, and rate limiting.

## COMPETITIONS

**Kaggle Competitions**

- **Santa 2025 - Christmas Tree Packing** — Rank: **425/3357** (**Top 13%**).
- **Predicting Student Test Scores** — Rank: **96/4317** (**Top 3%**).
- Kaggle profile: Ashraful

## PROJECTS

**AlzDetect-Ai: Explainable Alzheimer's Disease Classification System**
Developed an AI-powered system for multi-class Alzheimer's disease diagnosis from brain MRI scans using a custom multi-scale CNN and an ensemble of ResNet50 and Xception models. [GitHub] [Live Demo]

**TRIAGE: Risk-Controlled Semi-Supervised Learning Framework**
Engineered a pseudo-label admission mechanism in PyTorch for annotation-efficient retinal OCT classification, reducing error propagation under severe domain shifts. [GitHub] [arXiv]

**BeyondAccuracy: Anatomy-Guided Attribution Benchmark**
Implemented a pulmonary attribution quantification pipeline on chest X-rays to evaluate anatomical plausibility beyond standard accuracy metrics under cross-hospital domain shifts. [GitHub] [arXiv]

## SKILLS

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|---|---|
| **ML/DL Frameworks** | PyTorch, Torchvision, Hugging Face, MONAI, OpenCV, Scikit-learn |
| **Core Methodologies** | CNNs, Explainable AI (XAI), Semi-Supervised Learning, ViTs |
| **Programming Languages** | Python, C/C++, SQL, JavaScript |
| **Tools & Platforms** | Linux/Bash, Git, GitHub, Kaggle, LaTeX |

## REFERENCES

**Shyla Afroge**
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**Barshon Sen**
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